CAPSTONE – II
PROGRESS REPORT

Project Name: Sentiment Analysis using Hierarchical Reasoning Models and Mixture-of-Experts Architecture: Experimental Results and Analysis
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Tasks Completed:

1. **Deep Learning Infrastructure & Core Layers**
   - *Implemented a robust suite of configurable Neural Network layers in "**src/models/deep_leaning/models.py**":*
     - ***Normalization*:** Layer, Batch, Group, Instance, RMS, Weight, LocalResponse, and Lazy variants.
     - ***Pooling*:** Max, Average, Adaptive, Fractional, LP, and Unpooling.
     - ***Convolution*:** Standard 1D/2D/3D, Transposed, and Lazy convolutions.
     - ***Dropout*:** Standard, Alpha, and Feature Alpha dropout.
   - Developed ***DLModelLayers*** for dynamic, configuration-driven model instantiation.
   - Implemented Soft Dynamic Time Warping (**SoftDTW**) for differentiable sequence alignment and similarity measurement.
2. **Comprehensive Attention Mechanism Library**
   - *Implemented an extensive collection of Attention variants in "**src/models/deep_leaning/transformers/attention_units.py**":*
     - ***Efficiency-Focused***: FlashAttention, LinearAttention (Causal/Encoder/Decoder/Additive), SlimAttention, StrideSparseAttention, MultiQueryAttention, GroupedQueryAttention.
     - ***Structural & Graph***: GraphAttention, CrissCrossAttention, TriangularAttention, StructuredAttention (featuring diverse scoring modes: SSIM, Renyi, HSIC, CCA, Kernel-based, etc.).
     - ***Specialized***: MultiHeadedLatentAttention, LocationAttention, FAVOURPlusAttention, KAxAtAttention (Kolmogorov-Arnold inspired), XAttention, ChannelAttention.
   - These units support caching, various normalization strategies, and flexible integration into larger architectures.
3. **Large Language Model (LLM) Module Implementation**
   - Created ***LLMModule*** in "***src/models/deep_leaning/llm/llm_models.py***" to bridge the gap between local architecture and external pre-trained models.
   - Integration: Seamless support for HuggingFace ***transformers*** and ***sentence_transformers***.
   - ***Functionality:***
     - Robust model and tokenizer loading (Local fallback, remote download).
     - Flexible Embedding Generation with multiple pooling strategies (CLS, Mean, Max, Concatenation).
     - Built-in Dimensionality Reduction (PCA, t-SNE) and Clustering (K-Means, DBSCAN, Agglomerative) for latent space analysis.
4. **Clustering Algorithms**
   - *Implemented core clustering algorithms in "**src/models/deep_leaning/transformers/dummy.py**" (to be reorganized):*
     - K-Means, K-Medoids, K-Modes for diverse data types.

- CLARANS, Agglomerative (Hierarchical), Divisive Split, and BIRCH for scalable clustering.

Total Hours: 33 hours (1-week)

Plan for Next Week:

1. **Hierarchical Reasoning Model (HRM) Integration:**
   - Develop HRM modules to specifically address the low performance observed in the Medical domain.
   - Experiment with Mixture-of-Experts (MoE) to handle diverse dataset characteristics.
2. **Benchmarking & Optimization**
   - Evaluate the performance/latency trade-offs of the different Attention mechanisms (e.g., Flash vs. Linear vs. Structured).
   - Conduct ablation studies on the Medical Dataset using the new architecture to address previous performance gaps.
3. **Refinement**
   - Organize clustering algorithms into a dedicated module.
   - Finalize the Mixture-of-Experts (MoE) routing logic using the implemented soft-attention mechanisms.